Master of Computer Applications

Semester – III

Data Analytics for Business

Business Analytics Capstone Project

Title: Car Sales Business Analytics

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Car Sales Business Analytics

1. Problem Statement

A used-car business generates a large amount of vehicle sales data containing information about vehicle make, model, year, condition, mileage, market value, selling price, seller, state, and sale date. Analysing this data manually makes it difficult to identify sales trends, pricing patterns, high-performing vehicle brands, and seller or location performance.

The objective of this project is to develop an interactive **Car Sales Business Analytics Dashboard using Power BI**. The dashboard transforms raw car-sales data into meaningful KPIs, visualizations, and business insights that help users understand vehicle sales performance and support data-driven decision-making.

2. Objectives

- The main objectives of this project are:
- To analyse car sales data using Power BI.
- To clean and transform the raw dataset using Power Query.
- To prepare the data for business analysis.
- To create meaningful KPIs using DAX.
- To analyse selling price, market value, mileage, condition, make, state, seller, and sales trends.
- To create an interactive dashboard with charts, KPI cards, and slicers.
- To answer important business questions using the dashboard.
- To identify useful business insights and provide recommendations.

3. Methodology / Steps Followed

3.1 Data Collection

The car-sales dataset was collected and used as the main source for the project. The dataset contains vehicle details, market value, seller information, and final selling price.

The data was brought into the analytics workflow and prepared for Power BI analysis.

3.2 MySQL Database Connectivity

MySQL was used as part of the database connectivity workflow. The required database/table was connected and the car-sales data was made available for analysis.

The database connection allowed the data to be accessed in a structured format before performing the required data preparation and visualization.

3.3 Data Cleaning and Transformation

Power Query in Power BI was used for data cleaning and transformation.

The following activities were performed:

Reviewed the dataset columns.

Checked the data types of all columns.

Converted numerical fields into appropriate numeric data types.

Converted the sale date into Date/Time format.

Kept categorical fields such as make, model, body, state, transmission, and seller as text.

Checked for missing and blank values.

Removed or handled unsuitable records where required.

Prepared the cleaned data for analysis.

Ensured that selling price, MMR, odometer, year, and condition could be used correctly in calculations.

4. Data Modeling

The cleaned data was loaded into Power BI and structured for analysis.

The main car-sales data acts as the central table containing vehicle and transaction information.

The model was designed to allow Power BI visuals and DAX measures to work efficiently with filters and slicers.

5. DAX Measures and KPIs

DAX was used to create important measures for the dashboard.

Total Selling Price

Total Selling Price = SUM(CarSales[sellingprice])

Number of Vehicles Sold

Vehicles Sold = COUNTROWS(CarSales)

Average Selling Price

Average Selling Price = AVERAGE(CarSales[sellingprice])

Average MMR

Average MMR = AVERAGE(CarSales[mmr])

Average Odometer

Average Odometer = AVERAGE(CarSales[odometer])

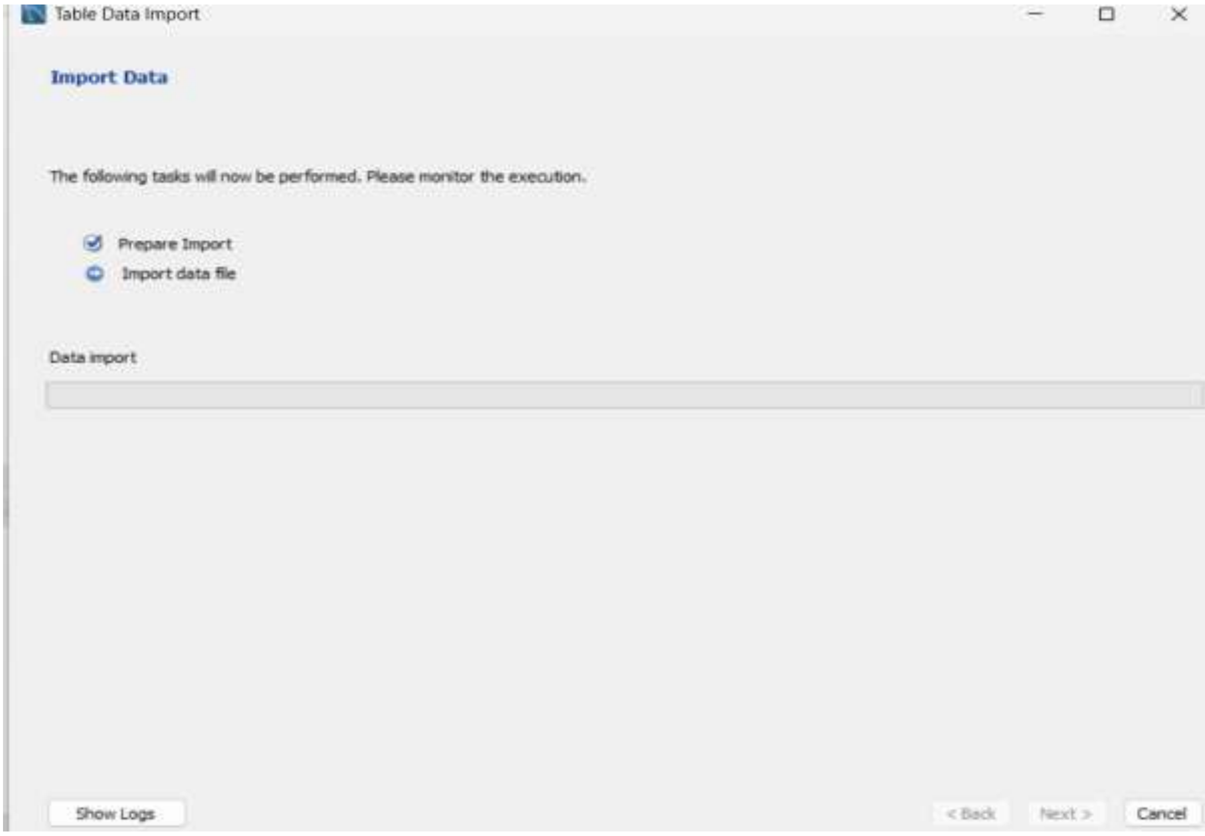
These measures were used in KPI cards and visualizations to provide a quick summary of overall sales performance.

6. Dashboard Pages / Screenshots

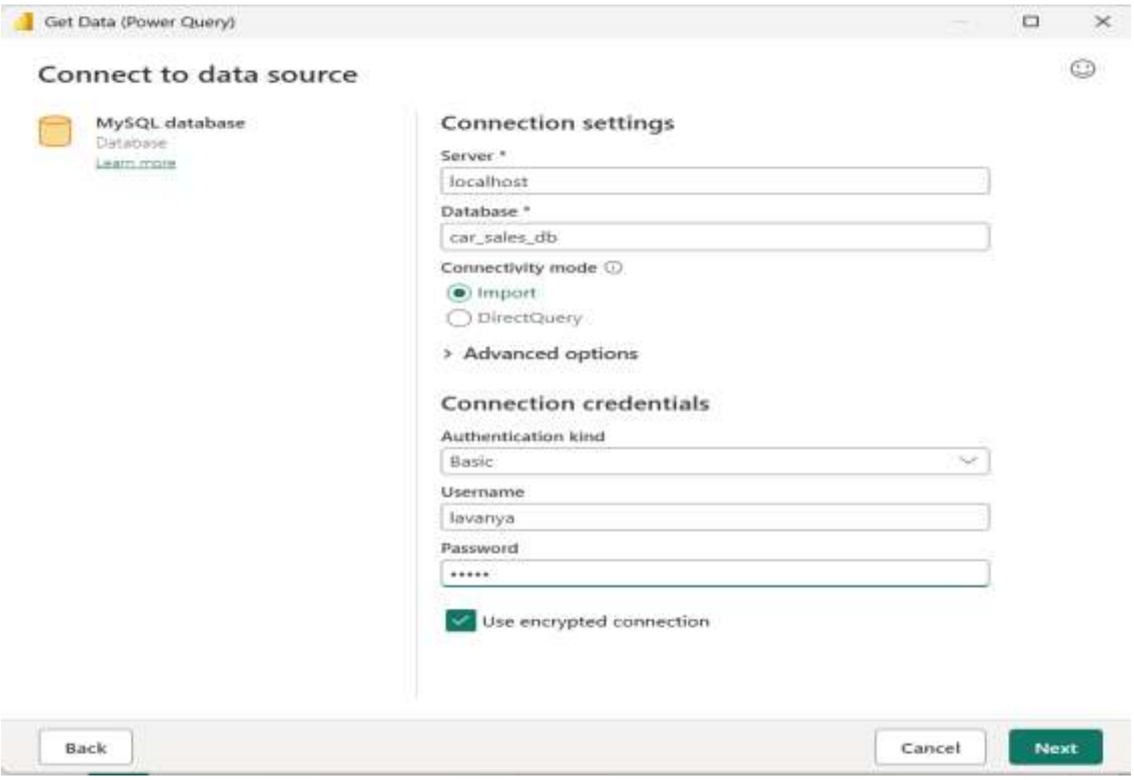
6.1 MySQL Database Creation

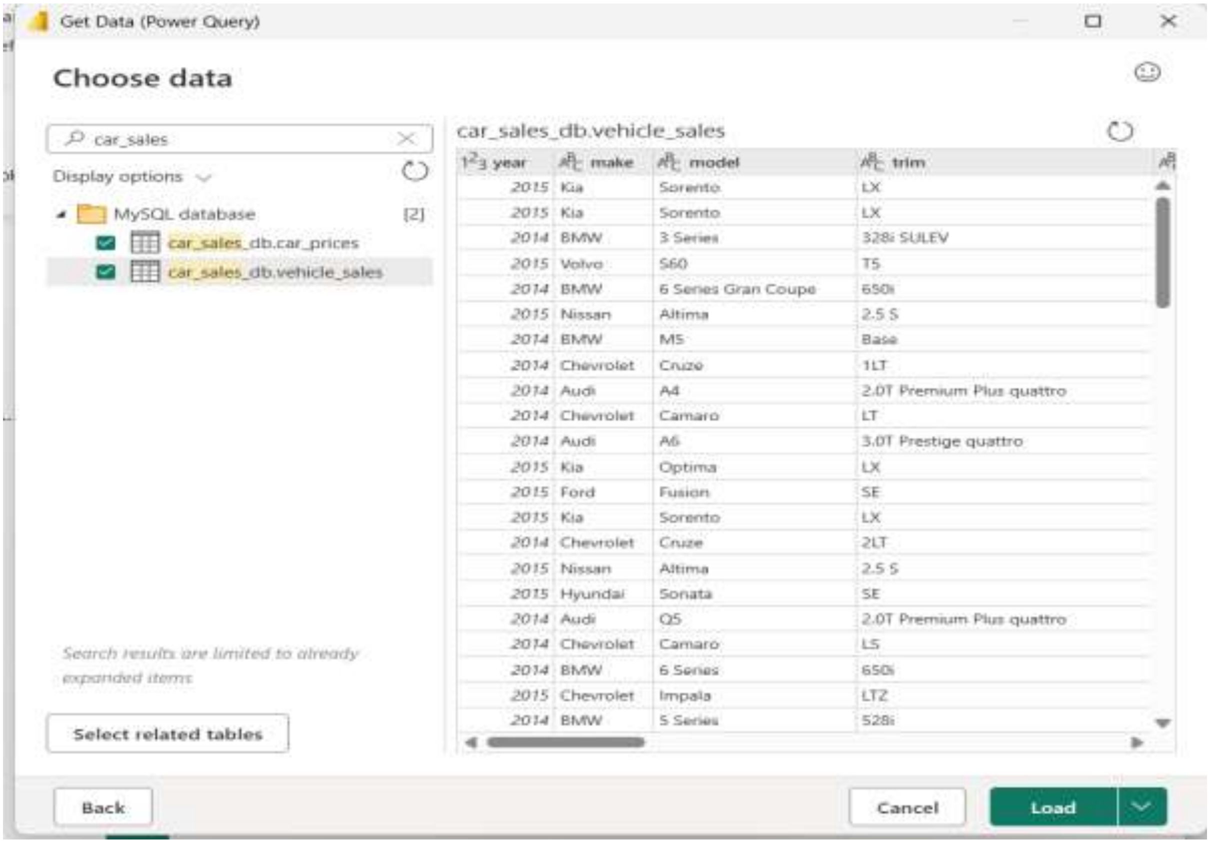
```
CREATE DATABASE car_sales_db;
USE car_sales_db;
SHOW DATABASES;

SELECT COUNT(*) AS total_records
FROM vehicle_sales;

SELECT *
FROM vehicle_sales
LIMIT 10;

SELECT COUNT(*) AS total_cars_sold
FROM vehicle_sales;

SELECT SUM(sellingprice) AS total_revenue
FROM vehicle_sales;

SELECT AVG(sellingprice) AS average_selling_price
FROM vehicle_sales;

```

```sql
SELECT
    seller,
    SUM(sellingprice) AS total_revenue
FROM vehicle_sales
GROUP BY seller
ORDER BY total_revenue DESC
LIMIT 10;

SELECT make, COUNT(*)
FROM vehicle_sales
GROUP BY make;

SELECT saledate
FROM vehicle_sales
LIMIT 10;

SELECT COUNT(*) AS total_records
FROM vehicle_sales;

SELECT *
FROM vehicle_sales
LIMIT 10;
```

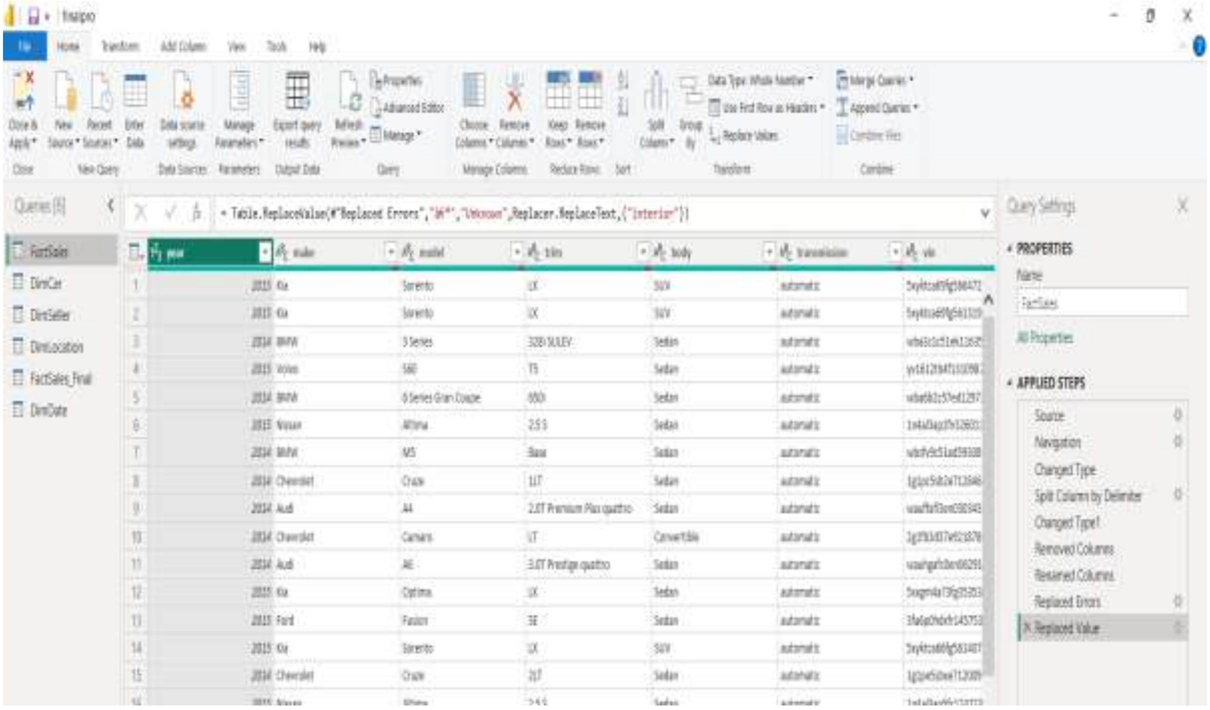


6.2 MySql DataBase Connect to Bower BI

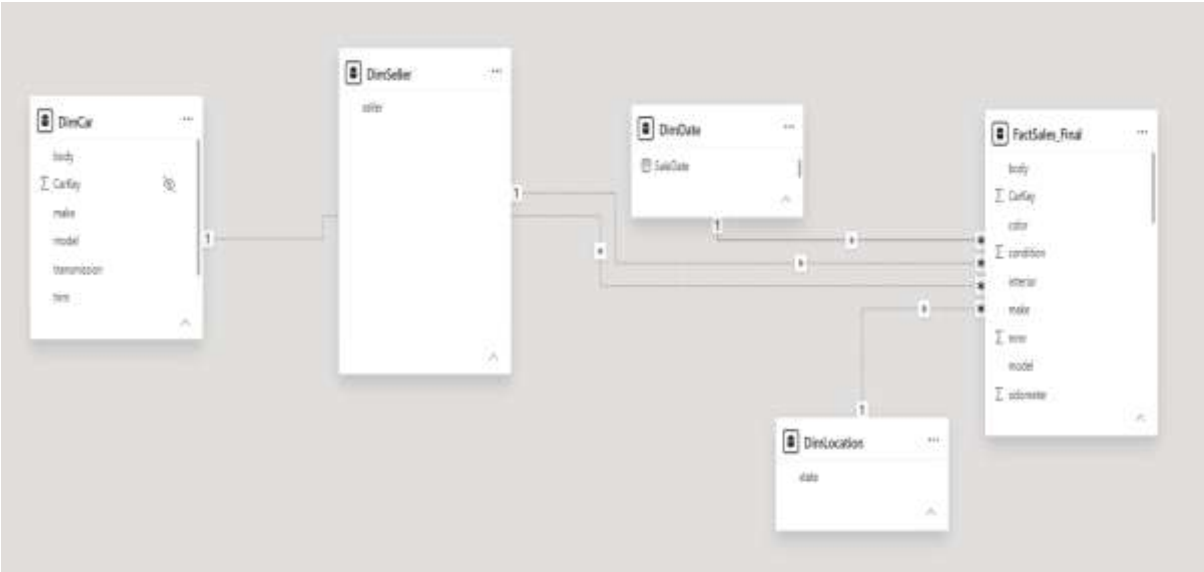




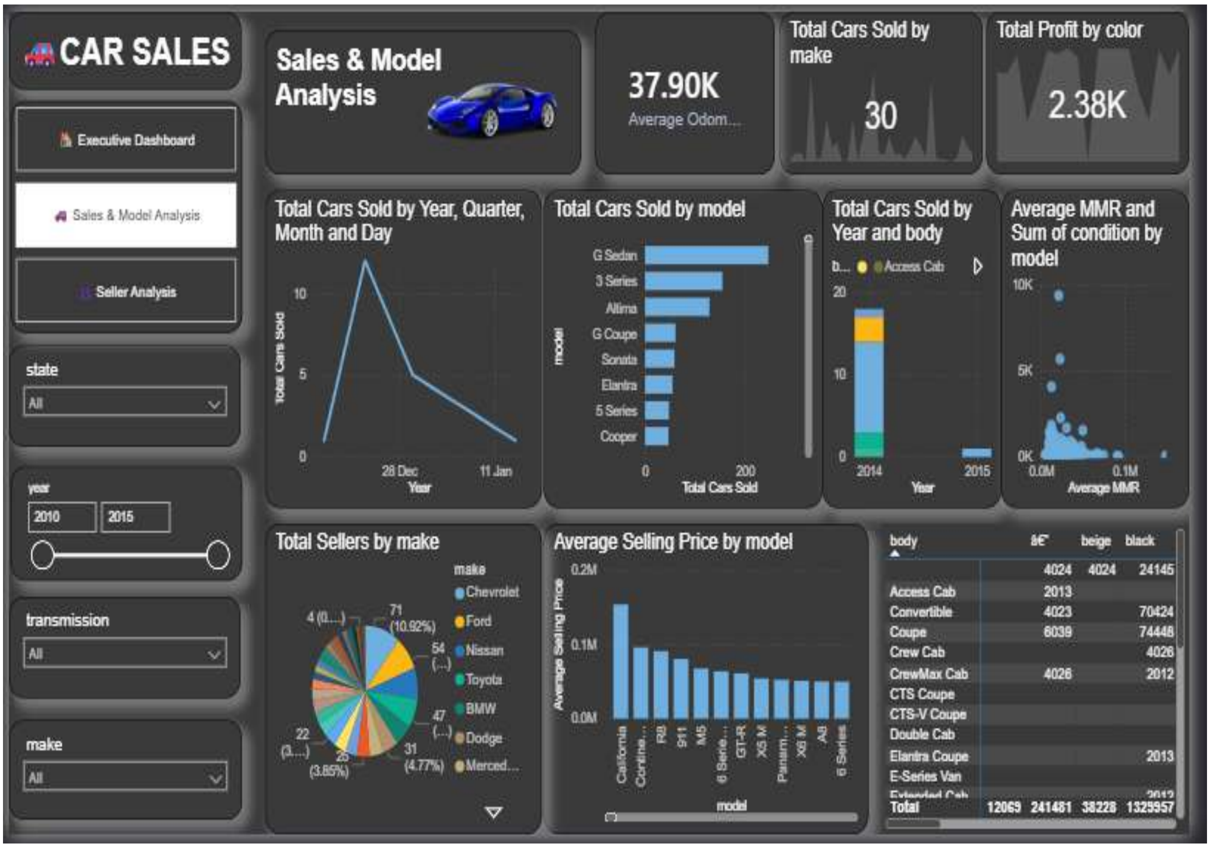
6.3 Data Transformation and Modeling

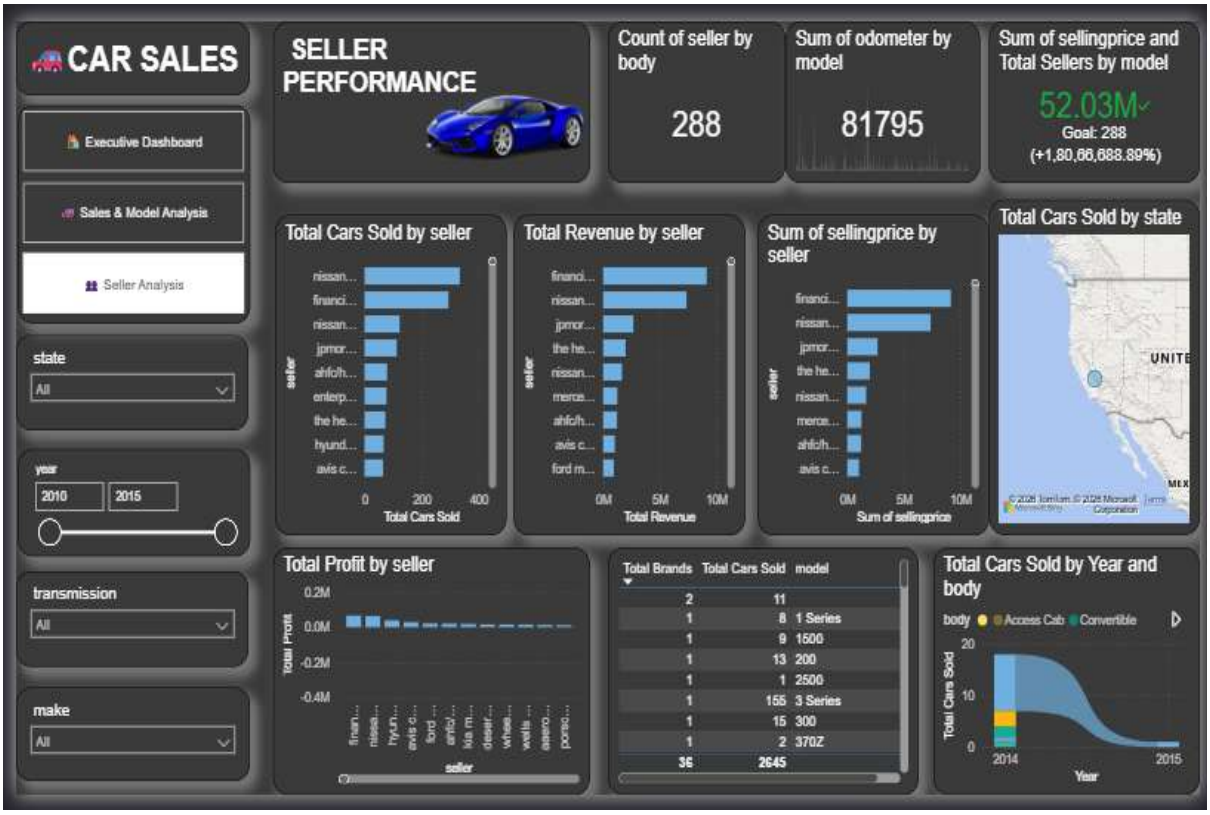


6.4 Data Base Schema



6.5 Dash Board Creation





7. Business Questions

The dashboard was developed to answer the following business questions:

What is the total selling value of the vehicles?

How many vehicles were sold?

What is the average selling price?

Which vehicle makes have higher sales performance?

Which vehicle body types have higher sales?

How do automatic and manual vehicles compare?

Which states have higher sales activity?

Which sellers have better sales performance?

How does mileage affect selling price?

How does selling price compare with MMR?

8. Output Results

- Interactive Car Sales Dashboard created using Power BI.
- Total selling price displayed through KPI.
- Total number of vehicles sold shown.
- Average selling price calculated.
- Average market value displayed.
- Average vehicle mileage analysed.
- Sales analysed by vehicle make.
- Sales analysed by state.
- Seller-wise sales performance analysed.
- Sales analysed by body type.
- Transmission-wise sales analysed.
- Selling price compared with MMR.
- Vehicle condition and mileage analysed.
- Sales trends visualized.
- Filters used to compare different vehicle and sales categories.

9. Business Insights

- The analysis provides several useful business insights.
- Vehicle Make
- Different vehicle manufacturers show different sales and pricing patterns. Comparing makes helps identify brands that contribute strongly to sales performance.
- Selling Price and MMR
- Comparing selling price with MMR helps identify whether vehicles are being sold above or below their estimated market value.
- Vehicle Condition
- Vehicle condition is an important factor in understanding differences in selling prices. Vehicles with better condition can have stronger selling prices.
- Odometer
- Mileage provides useful information about vehicle usage. Vehicles with higher mileage may have different selling-price patterns compared with vehicles having lower mileage.
- State
- State-wise analysis helps identify locations with higher sales activity and allows businesses to compare regional performance.
- Seller

- Seller-wise analysis helps identify sellers with stronger sales performance and provides an opportunity to compare different selling channels.
- Body Type and Transmission
- Body type and transmission are useful categories for understanding customer demand and comparing sales performance between vehicle segments.

10. Business Recommendations

Based on the analysis, the following recommendations can be made:

Use MMR, vehicle condition, and mileage together when evaluating vehicle prices.

Monitor high-performing vehicle makes and models.

Maintain suitable inventory for vehicle categories with strong sales performance.

Analyse state-wise performance to identify strong and weak markets.

Monitor seller performance regularly.

Use the dashboard for regular sales and pricing monitoring.

Compare actual selling prices with MMR before making pricing decisions.

Consider vehicle condition and mileage when estimating expected selling prices.

Use interactive dashboard filters to support faster business analysis.

Update the dashboard regularly when new sales data becomes available.

11. Power BI Service and Refresh Procedure

The completed Power BI report can be published to Power BI Service for online access and future data refresh.

The procedure is:

Open the completed Power BI Desktop report.

Save the final .pbix file.

Select **Publish** from Power BI Desktop.

Select the required Power BI workspace.

Open the published report in Power BI Service.

Check the published report and dataset.

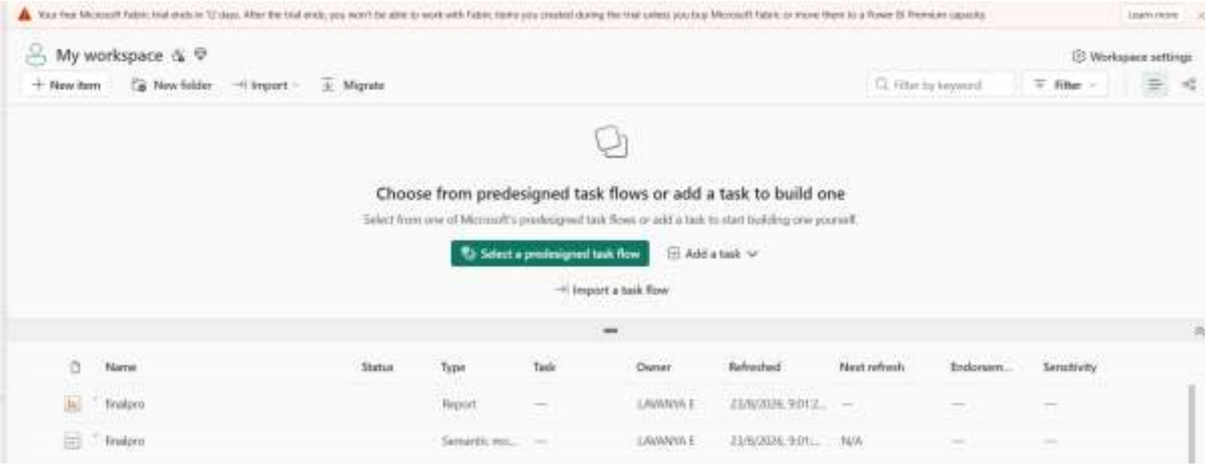
Configure the required data-source credentials.

Configure scheduled refresh if required.

Run a manual refresh to verify the connection.

Check the refresh history to confirm successful refresh.

Power BI Service Screenshot:



Refresh History Screenshot:

12. Conclusion

The **Car Sales Business Analytics Project** successfully demonstrates an end-to-end data analytics workflow using Power BI.

The raw car-sales dataset was cleaned and transformed using Power Query and prepared for business analysis. DAX measures were created to calculate important KPIs such as total selling price, number of vehicles sold, average selling price, average MMR, and average odometer.

An interactive Power BI dashboard was developed to analyse vehicle sales based on make, model, state, seller, body type, transmission, condition, mileage, market value, selling price, and sale date.

The dashboard provides a simple and effective way to identify sales patterns, compare vehicle categories, analyse pricing, and support data-driven business decisions.

Overall, the project demonstrates how raw business data can be transformed into meaningful visual insights using modern business intelligence tools.

13. Learning Outcomes

- Worked with real-world business data.
- Cleaned and transformed data using Power Query.
- Created data models in Power BI.
- Created DAX measures and KPIs.
- Designed interactive Power BI dashboards.
- Created different data visualizations.
- Analysed business data and identified insights.
- Made data-based business recommendations.
- Learned Power BI Service publishing and refresh.
- Improved business analytics and reporting skills.

Github link: https://github.com/lavanyae4319/Data_Analytics_Assignments